Networking concepts and security

Prof .Vaishalee Joishar
CSE-Cyber Security
PIET,Parul University

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Unit - 1

Types of Network & OSI Layers

Typess of Network

PAN (Personal Area Network)
LAN (Local Area Network)
MAN (Metropolitan Area Network)
WAN (Wide Area Network)

7 types of networks

	SIZE	DEVICES	CONNECTIVITY
PAN Devices connected around an individual	Small area of coverage; usually around 10 meters	IoT devices, tablets, wearables	Bluetooth, Zigbee, USB
LAN Devices connected to each other in one physical location	Up to hundreds of meters	Computers, smartphones, access points, routers, switches	Fiber optics, Ethernet, wireless, cellular
MAN Several LANs connected in a municipality	Up to 50 km	Computers, smartphones, access points, routers, switches	Fiber optics, Ethernet, wireless, cellular
Campus Several LANs connected in a general area	Between 1 km and 5 km	Computers, smartphones, access points, routers, switches	Fiber optics, Ethernet, wireless, cellular
WAN Connection of LANs linked around the world	Global	Computers, gateways, routers, switches	Fiber optics, Ethernet, wireless, cellular
CDN Connection of servers linked around the world to distribute rich media content	Global	Proxy servers, origin servers	Points of presence
VPN Virtual private network overlay on an existing network	Global	Computers, smartphones, tablets	Virtual connections with tunneling protocols

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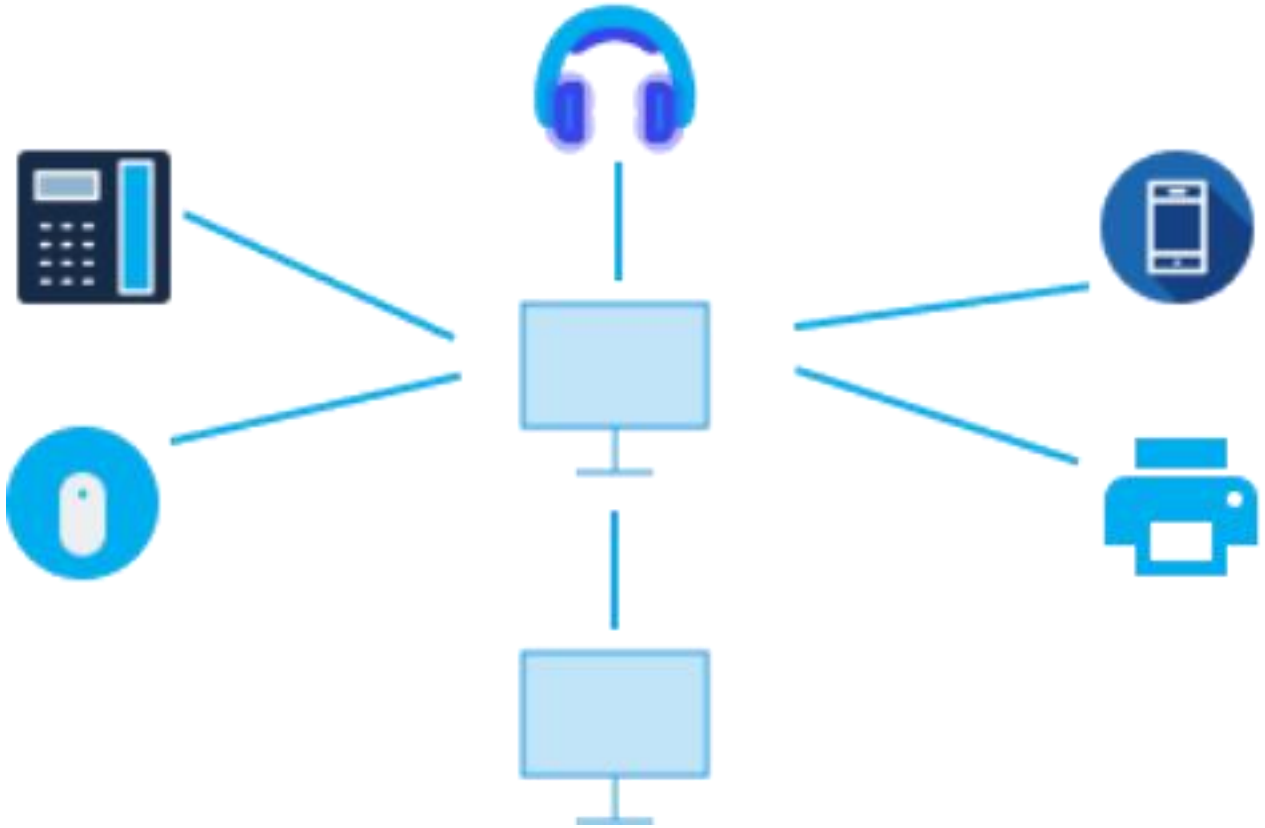
PAN (Personal Area Network)

A personal area network is a computer network for interconnecting electronic devices within an individual person's workspace.

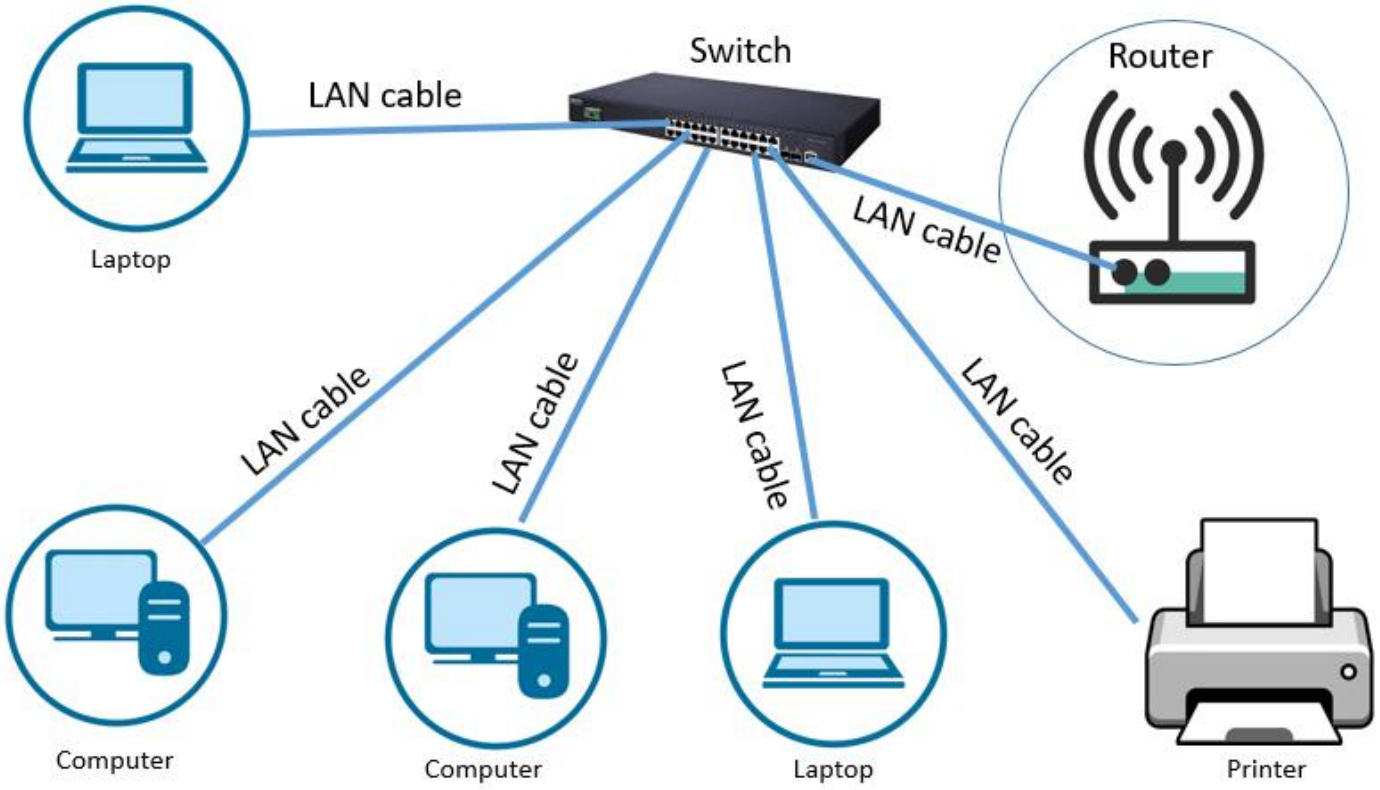
A PAN provides data transmission among devices such as computers, smartphones, tablets and personal digital assistants.

Personal Area Network (PAN)



LAN (Local Area Network)

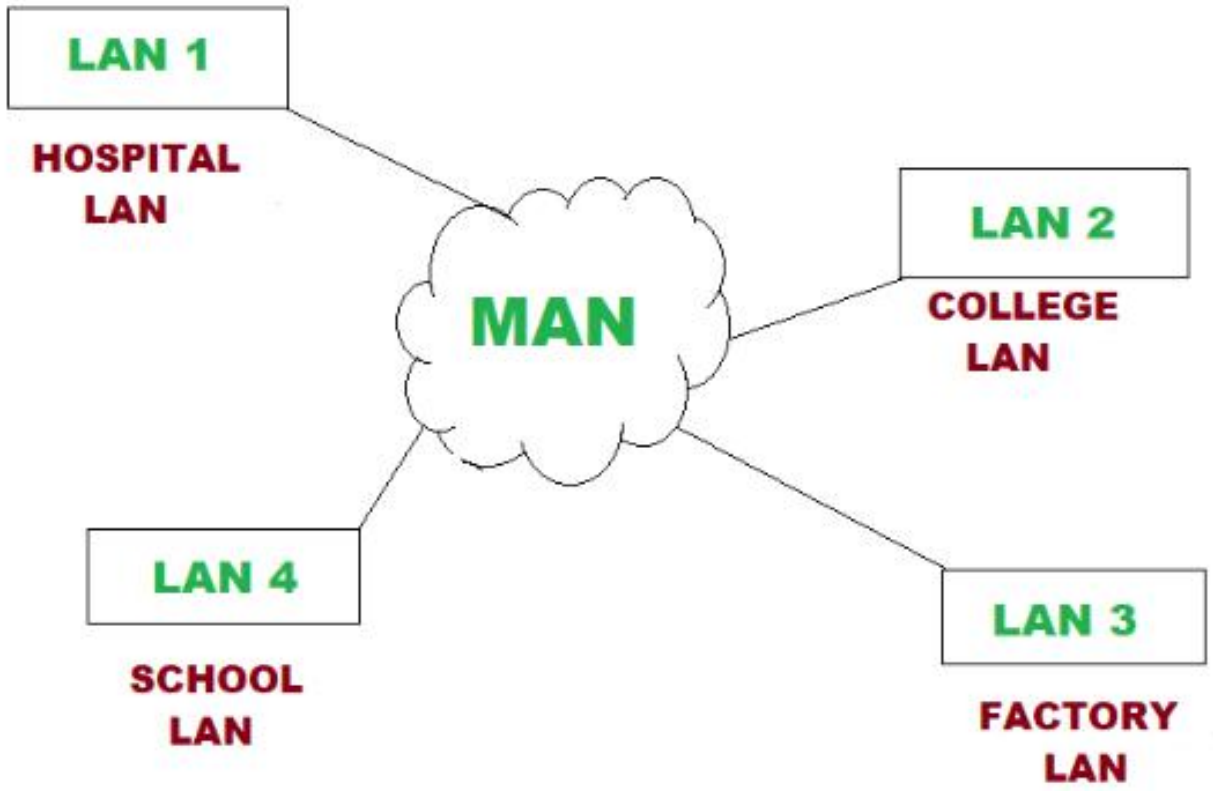
A local area network is a computer network that interconnects computers within a limited area such as a residence, school, laboratory, university campus or office building.



Local Area Network

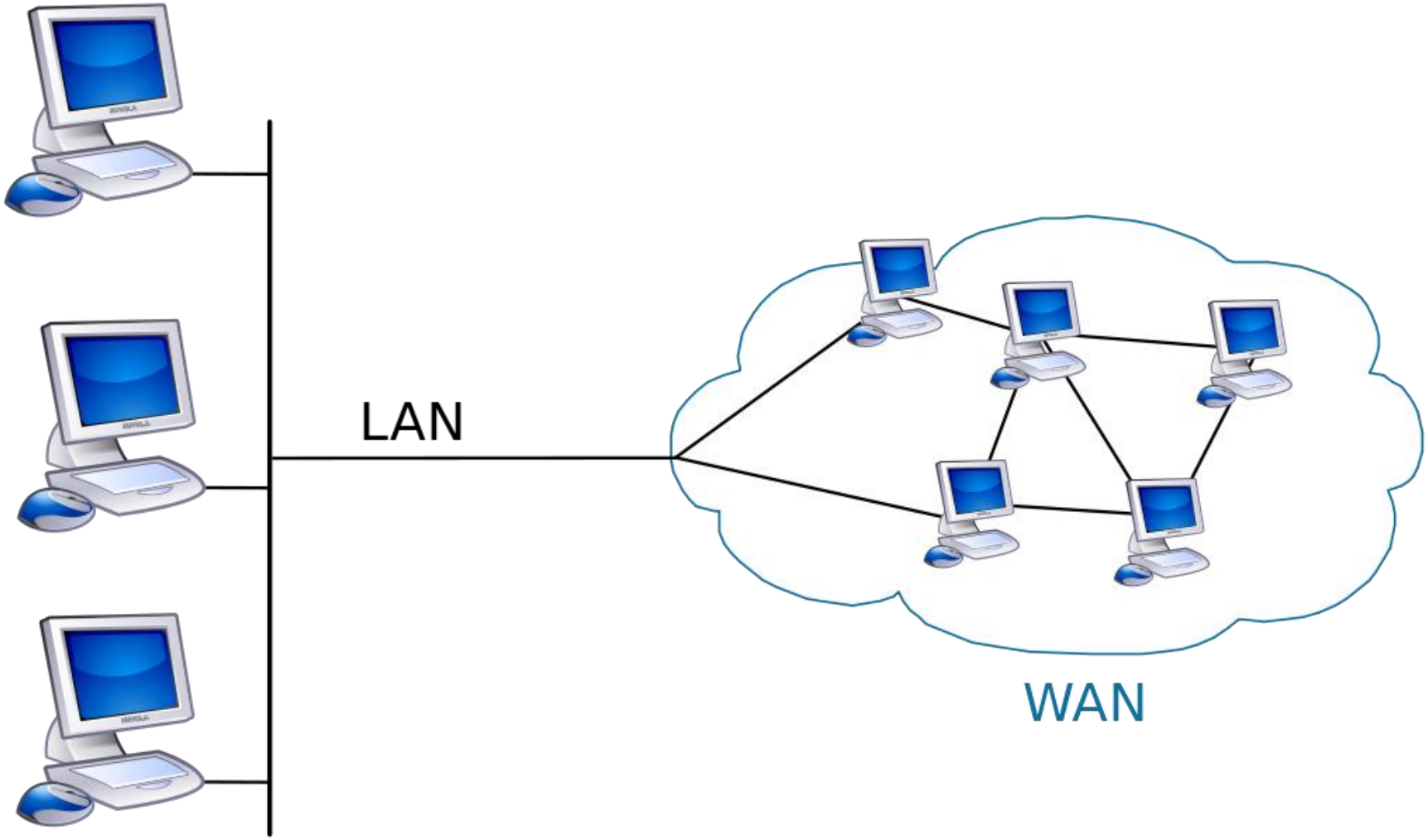
Metropolitan Area Network

A metropolitan area network is a computer network that interconnects users with computer resources in a geographic region of the size of a metropolitan area.


LAN 1
HOSPITAL LAN
LAN 2
COLLEGE LAN
MAN
LAN 4
SCHOOL LAN
LAN 3
FACTORY LAN

Wide Area Network

A wide area network is a telecommunications network that extends over a large geographic area. Wide area networks are often established with leased telecommunication circuits


LAN
WAN

IP address

IP stands for "Internet Protocol"
IP address is a unique address that identifies a device on the internet or a local network.
IP address is a string of numbers separated by dot.
IP addresses are expressed as a set of four numbers
Each number in the set can range from 0 to 255.
IP addresses are not random. They are mathematically produced and allocated by the Internet Assigned Numbers Authority (IANA)
Ex.:- 192.158.1.38

172 . 16 . 254 . 1

10101100 .00010000 .11111110 .00000001

1 byte=8 bits

32 bits (4 x 8), or 4 bytes

There are two versions of IP address
IP version 4 (IPv4)
IP version 6 (IPv6)

IP version 4 (IPv4)
IPv4 has a 32-bit address length.
It Supports Manual and DHCP address configuration
It can generate 4.29×109 address space

The Security feature is dependent on application
Address representation of IPv4 is in decimal
In IPv4 Encryption and Authentication facility not provided
IPv4 has a header of 20-60 bytes
Ex:- 192.168.1.165

IP version 6 (IPv6)
IPv6 has a 128-bit address length
It supports Auto and renumbering address configuration
Address space of IPv6 is quite large it can produce
3.4×1038 address space
IPSEC is an inbuilt security feature in the IPv6 protocol
Address Representation of IPv6 is in hexadecimal
In IPv6 Encryption and Authentication are provided
IPv6 has header of 40 bytes fixed
Ex:- 2001:0db8:0000:0000:0000:ff00:0042:7879

Class A Network

This IP address class is used when there are a large number of hosts. In a Class A type of network, the first 8 bits (first octet) identify the network, and the remaining have 24 bits for the host into that network.

An example of a Class A address is 102.168.212.226. Here, “102” helps you identify the network and 168.212.226 identify the host.

Address Range:- 1 to 126

Class B Network

In class B the first 16 bits (first two octets) help you identify the network. The other remaining 16 bits indicate the host within the network.

An example of Class B IP address is 168.212.226.204, where *168 212* identifies the network and *226.204* helps you identify the Hut network host.

Address Range:- 128 to 191

Class C Network

Class C is a type of IP address that is used for the small network. In this class, three octets are used to indent the network.

In this type of network addressing method, the first two bits are set to be 1, and the third bit is set to 0, which makes the first 24 bits of the address them and the remaining bit as the host address. Mostly local area network used Class C IP address to connect with the network.

Address Range:- 192 to 223

Class D Network

Class D addresses are only used for multicasting applications. Class D is never used for regular networking operations. This class addresses the first three bits set to “1” and their fourth bit set to use for “0”. Class D addresses are 32-bit network addresses. All the values within the range are used to identify multicast groups uniquely.

Therefore, there is no requirement to extract the host address from the IP address, so Class D does not have any subnet mask.

Address Range:- 224 to 239

Class E Network

Class E IP address is defined by including the starting four network address bits as 1, which allows you two to incorporate addresses from 240.0.0.0 to 255.255.255.255.

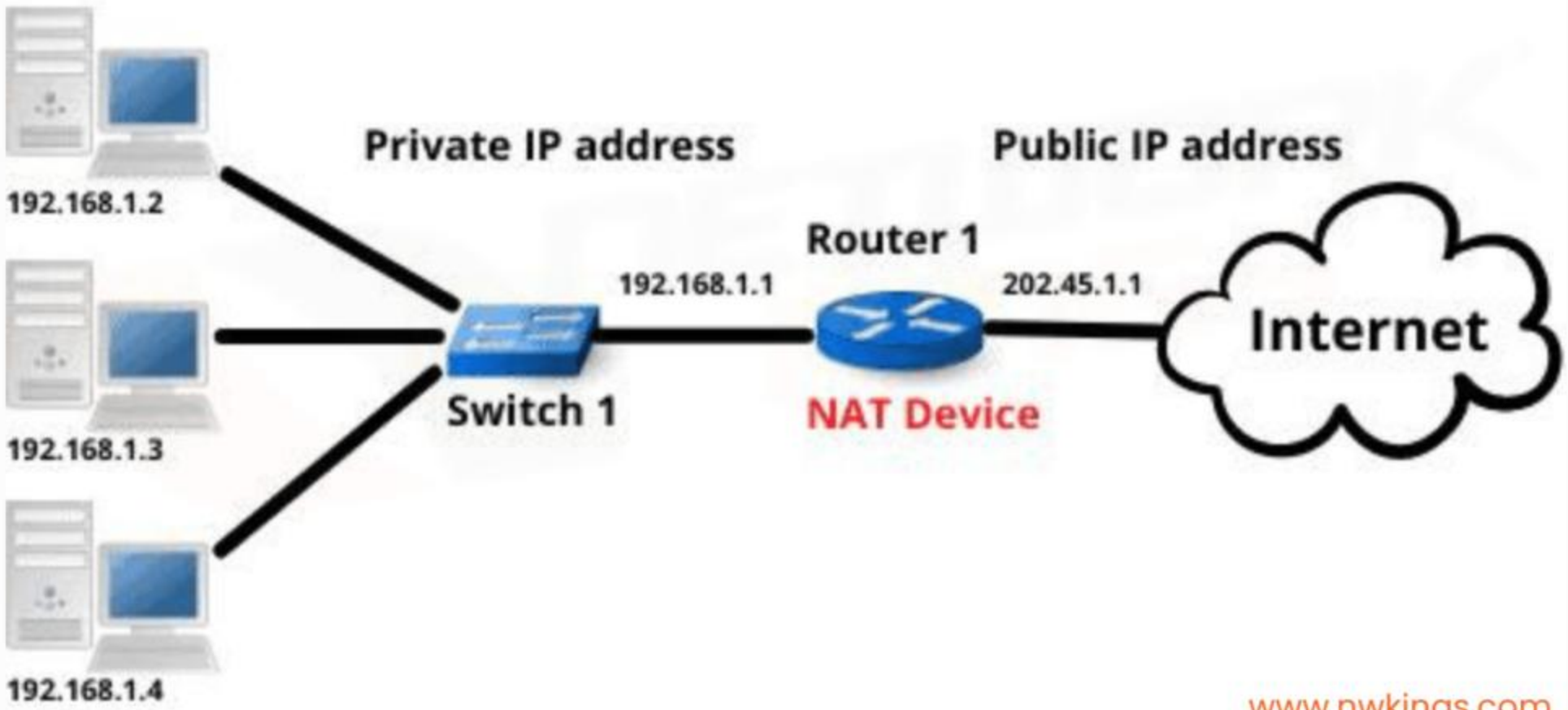
E class is reserved for research and Development Purposes, and its usage is never defined. Therefore, many network implementations discard these addresses as undefined or illegal.

Address Range:- 240 to 254

Network Address Translation(NAT)

NAT stands for network address translation.
Network Address Translation (NAT) is a process in which one or more local IP address is translated into one or more Global IP address and vice versa in order to provide Internet access to the local hosts.

Network Address Translation



www.nwkings.com

Network Address Translation working

The router which has one interface in the local (inside) network and one interface in the global (outside) network. When a packet traverse outside the local (inside) network, then NAT converts that local (private) IP address to a global (public) IP address. When a packet enters the local network, the global (public) IP address is converted to a local (private) IP address.

If NAT runs out of addresses, no address is left in the pool configured then the packets will be dropped and an Internet Control Message Protocol (ICMP) host unreachable packet to the destination is sent.

Why mask port numbers?

Suppose, in a network, two hosts A and B are connected. Now, both of them request for the same destination, on the same port number, say 1000, on the host side, at the same time.

If NAT does only translation of IP addresses, then when their packets will arrive at the NAT, both of their IP addresses would be masked by the public IP address of the network and sent to the destination.

Destination will send replies to the public IP address of the router.

Inside local address – An IP address that is assigned to a host on the Inside (local) network. The address is probably not an IP address assigned by the service provider i.e., these are private IP addresses. This is the inside host seen from the inside network.

Inside global address – IP address that represents one or more inside local IP addresses to the outside world. This is the inside host as seen from the outside network.

Outside local address – This is the actual IP address of the destination host in the local network after translation.

Outside global address – This is the outside host as seen from the outside network. It is the IP address of the outside destination host before translation.

Advantages of NAT

NAT conserves legally registered IP addresses.

It provides privacy as the device's IP address, sending and receiving the traffic, will be hidden.

Eliminates address renumbering when a network evolves.

Disadvantage of NAT

Translation results in switching path delays.

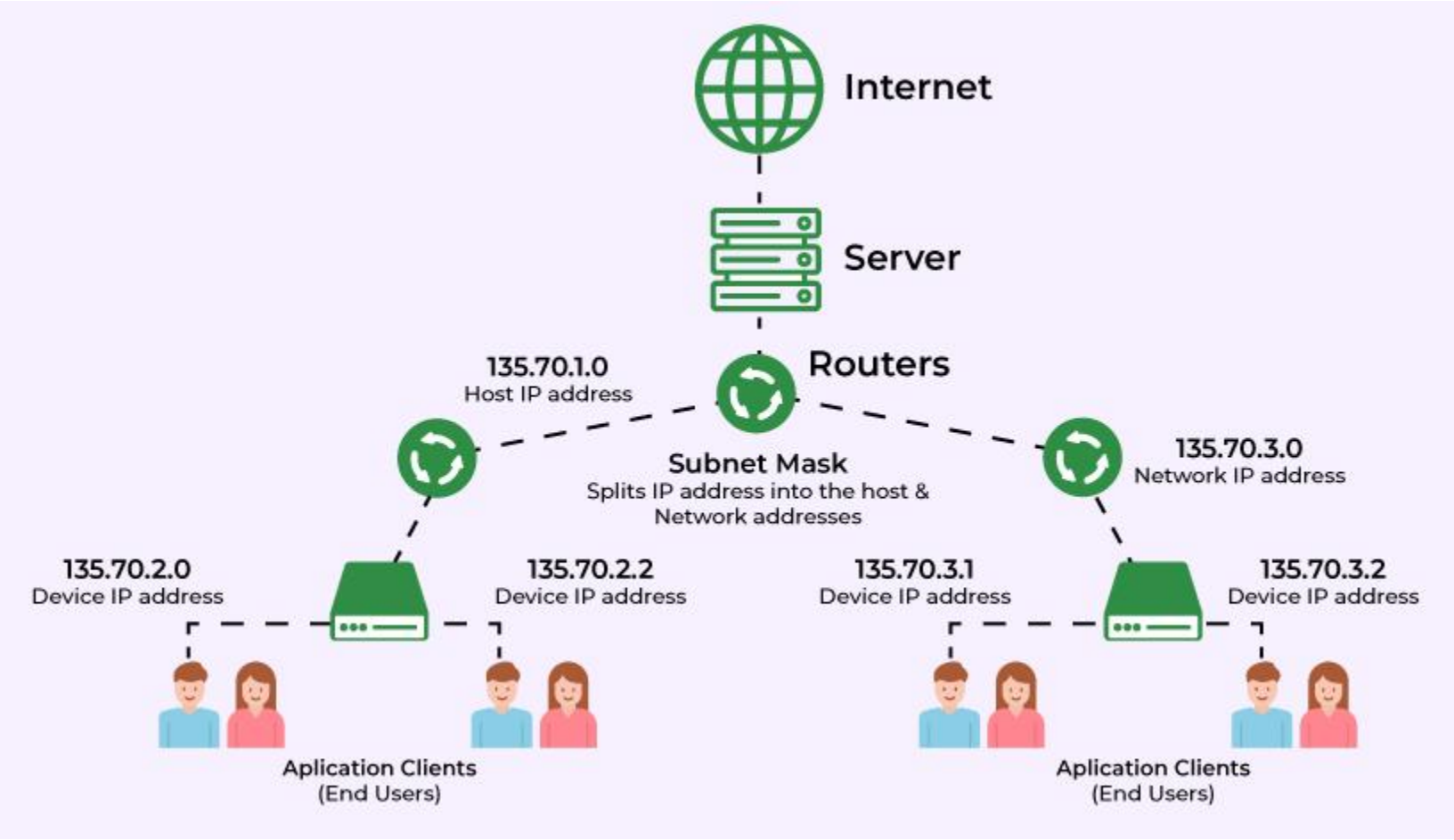
Certain applications will not function while NAT is enabled.

Complicates tunneling protocols such as IPsec.

Also, the router being a network layer device, should not tamper with port numbers(transport layer) but it has to do so because of NAT.

IP subnets

In networking, an IP subnet, or simply subnet, is a logical subdivision of an IP network. Subnetting is the process of dividing a larger IP network into smaller, more manageable sub-networks. This is done for various reasons, including improving performance, enhancing security, and facilitating efficient use of IP addresses.


Internet
Server
Routers
135.70.1.0
Host IP address
Subnet Mask
Splits IP address into the host &
Network addresses
135.70.3.0
Network IP address
135.70.2.0
Device IP address
135.70.2.2
Device IP address
135.70.3.1
Device IP address
135.70.3.2
Device IP address
Aplication Clients
(End Users)
Aplication Clients
(End Users)

Key concepts related to IP subnets:

IP Address:

An IP address is a numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication. IPv4 addresses are typically written in the form of four sets of decimal numbers separated by dots (e.g., 192.168.1.1).

Key concepts related to IP subnets:

Subnet Mask:

The subnet mask is a 32-bit number that divides an IP address into network and host portions. It consists of a series of contiguous '1' bits followed by a series of contiguous '0' bits. For example, the subnet mask 255.255.255.0 (or /24 in CIDR notation) means that the first 24 bits of the IP address represent the network, and the remaining 8 bits represent the host.

CIDR Notation:

CIDR (Classless Inter-Domain Routing) notation is a way to represent IP addresses and their associated routing prefix. It allows for a flexible and efficient allocation of IP addresses. For example, instead of using traditional subnet masks, CIDR notation might express a subnet as 192.168.1.0/24.

Subnetting:

Subnetting involves dividing an IP network into sub-networks to improve performance and security. It allows for more efficient use of IP addresses by allocating them to smaller groups within the larger network. Subnetting is often used in conjunction with CIDR notation.

Private and Public IP Addresses:

IP addresses are categorized as private or public. Private IP addresses are reserved for use within a private network and are not routable on the public Internet. Public IP addresses are assigned by Internet authorities and are routable on the Internet.

VLSM (Variable Length Subnet Masking):
VLSM is a technique that allows for using different subnet masks within the same network address space. This enables more efficient use of IP addresses by assigning larger subnets to network segments with more hosts and smaller subnets to segments with fewer hosts.

DHCP Server

Dynamic Host Configuration Protocol is an application layer protocol
which is used to provide:
Subnet Mask (Option 1 – e.g., 255.255.255.0)
Router Address (Option 3 – e.g., 192.168.1.1)
DNS Address (Option 6 – e.g., 8.8.8.8)
Vendor Class Identifier (Option 43 – e.g., ‘unifi’ = 192.168.1.9
##where unifi = controller)

Subnet Mask

A subnet mask is a 32-bit number created by setting host bits to all 0s and setting network bits to all 1s. In this way, the subnet mask separates the IP ...

A subnet mask is used to divide an IP address into two parts. One part identifies the host (computer), the other part identifies the network to which it belongs.

Vendor Class Identifier

Using the vendor class identifier allows DHCP administrators to assign vendor-specific DHCP options to devices without running the risk of duplicating options within the DHCP scope.

Advantages –

centralized management of IP addresses

Ease of adding new clients to a network

Reuse of IP addresses reducing the total number of IP addresses that are required

Simple reconfiguration of the IP address space on the DHCP server without needing to reconfigure each client

The DHCP protocol gives the network administrator a method to configure the network from a centralized area.

With the help of DHCP, easy handling of new users and reuse of IP address can be achieved.

Ports

A port is a virtual point where network connections start and end.

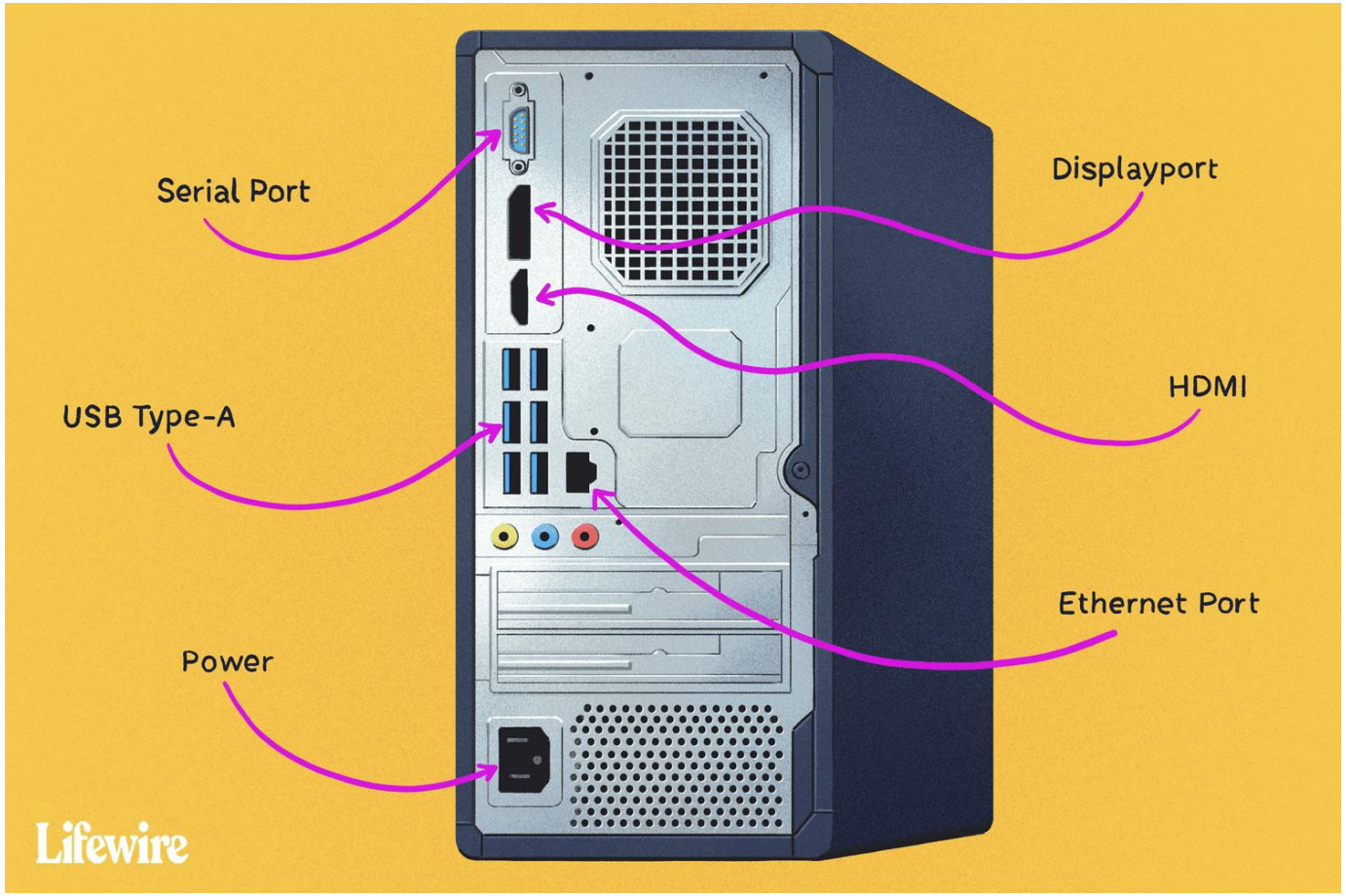
Ports are software-based and managed by a computer's operating system.

Each port is associated with a specific process or service.

A port is a virtual point where network connections start and end. Ports are software-based and managed by a computer's operating system.

Each port is associated with a specific process or service.

Ports allow computers to easily differentiate between different kinds of traffic


Serial Port
Displayport
HDMI
USB Type-A
Ethernet Port
Power
Lifewire

Port number

Ports are standardized across all network-connected devices, with each port assigned a number.
Most ports are reserved for certain protocols — for example, all Hypertext Transfer Protocol (HTTP) messages go to port 80.
While IP addresses enable messages to go to and from specific devices, port numbers allow targeting of specific services or applications within those devices.

Ports are a transport layer (layer 4) concept.
Only a transport protocol such as the Transmission Control Protocol (TCP) or User Datagram Protocol (UDP) can indicate which port a packet should go to.
TCP and UDP headers have a section for indicating port numbers.
Network layer protocols — for instance, the Internet Protocol (IP) — are unaware of what port is in use in a given network connection.
In a standard IP header, there is no place to indicate which port the data packet should go to.

There are 65,535 possible port numbers, although not all are in common use. Some of the most commonly used ports, along with their associated networking protocol, are:

There are 65,535 possible port numbers, although not all are in common use. Some of the most commonly used ports, along with their associated networking protocol, are:
Ports 20 and 21: File Transfer Protocol (FTP). FTP is for transferring files between a client and a server.

Port 22: Secure Shell (SSH). SSH is one of many tunneling protocols that create secure network connections.

Port 25: Simple Mail Transfer Protocol (SMTP). SMTP is used for email.

Port 53: Domain Name System (DNS). DNS is an essential process for the modern Internet; it matches human-readable domain names to machine-readable IP addresses, enabling users to load websites and applications without memorizing a long list of IP addresses.

Port 80: Hypertext Transfer Protocol (HTTP). HTTP is the protocol that makes the World Wide Web possible.

Port 123: Network Time Protocol (NTP). NTP allows computer clocks to sync with each other.

Port 179: Border Gateway Protocol (BGP). BGP is essential for establishing efficient routes between the large networks that make up the Internet (these large networks are called autonomous systems). Autonomous systems use BGP to broadcast which IP addresses they control

Port 443: HTTP Secure (HTTPS). HTTPS is the secure and encrypted version of HTTP. All HTTPS web traffic goes to port 443. Network services that use HTTPS for encryption, such as DNS over HTTPS, also connect at this port.

Port Range Groups

0 to 1023 – Well known port numbers. Only special companies like Apple QuickTime, MSN, SQL Services, Gopher Services, and other prominent services have these port numbers.

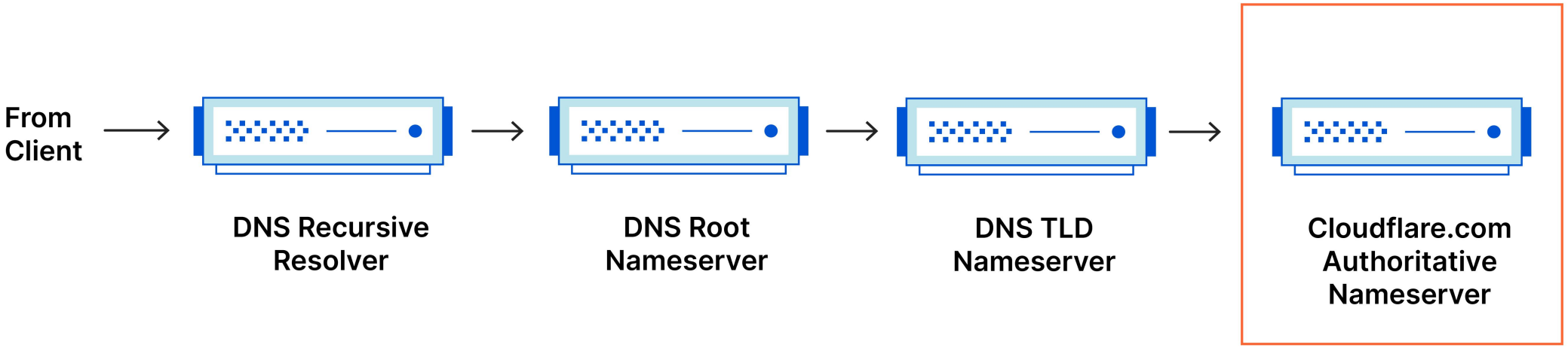
1024 to 49151 – Registered ports; meaning they can be registered to specific protocols by software corporations.

49152 to 65536 – Dynamic or private ports; meaning that they can be used by just about anybody.

DNS (Domain Name System)

DNS, or Domain Name System, is a critical component of computer networks that translates human-readable domain names into IP addresses. Instead of relying on users to remember and use numerical IP addresses, DNS provides a hierarchical and distributed system that allows users to use easily memorable domain names to access resources on the Internet.

DNS Record Request Sequence



Key aspects of DNS in networks:

Domain Names:
A domain name is a human-readable label assigned to an IP address or a set of IP addresses. For example, www.example.com is a domain name. Domain names are organized hierarchically, with the top-level domain (TLD) at the rightmost part (e.g., .com, .org, .net).

DNS Servers:

DNS operates in a distributed manner through a network of DNS servers. These servers are categorized into several types:

Root DNS Servers: These servers are at the top of the DNS hierarchy and provide information about the authoritative DNS servers for top-level domains.

Top-Level Domain (TLD) DNS Servers: These servers handle requests for specific TLDs (e.g., .com, .org).

Authoritative DNS Servers: These servers store and provide information about a specific domain. There are authoritative servers for each domain.

Recursive DNS Servers: These servers perform the actual work of translating domain names to IP addresses. They can query other DNS servers to resolve a domain name.

DNS Resolution Process:

When a user types a domain name into a web browser, the following steps occur:

The local device checks its local DNS cache to see if it already has the IP address for the requested domain.

If not, the request is sent to a recursive DNS server (often provided by the Internet Service Provider). This server may have the information in its cache or may need to query other DNS servers to resolve the domain.

The recursive server checks the root DNS servers to find the TLD DNS servers responsible for the domain.

The TLD DNS server directs the recursive server to the authoritative DNS server for the specific domain.
The authoritative DNS server provides the IP address for the requested domain back to the recursive server.
The recursive server caches this information and returns the IP address to the user's device, which can then access the desired resource.

DNS Records:

DNS servers store various types of records associated with domain names. Common DNS record types include:
A Record: Maps a domain name to an IPv4 address.
AAAA Record: Maps a domain name to an IPv6 address.
CNAME (Canonical Name) Record: Alias of one domain to another.
MX (Mail Exchange) Record: Specifies mail servers responsible for receiving email for the domain.
NS (Name Server) Record: Specifies authoritative DNS servers for the domain.

Proxy Servers

The proxy server is a computer on the internet that accepts the incoming requests from the client and forwards those requests to the destination server.

It works as a gateway between the end-user and the internet. It has its own IP address. It separates the client system and web server from the global network.

Proxy Servers

we can say that the proxy server allows us to access any websites with a different IP address. It plays an intermediary role between users and targeted websites or servers.

It collects and provides information related to user requests. The most important point about a proxy server is that it does not encrypt traffic.

Proxy Server List
User
Proxy Server
Internet

The proxy server accepts the request from the client and produces a response based on the following conditions:
If the requested data or page already exists in the local cache, the proxy server itself provides the required retrieval to the client.

If the requested data or page does not exist in the local cache, the proxy server forwards that request to the destination server.

The proxy servers transfer the replies to the client and also being cached to them.
Therefore, it can be said that the proxy server acts as a client as well as the server.

Communication without proxy server

Communication with proxy server

VPN	VS	PROXY
Encrypts the whole internet connection		Must be configured for each app (e.g Chrome, BitTorrent)
Changes your IP		Changes your IP
Has its own software		Doesn't have its own software
Encrypts the data being transferred		Doesn't encrypt the data being transferred
Slower than a proxy		Faster than a VPN
More costly than a proxy		Cheaper than a VPN

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DNS (Domain Name System) server

A DNS (Domain Name System) server is a critical component of the internet infrastructure that translates human-readable domain names into IP addresses. This process is essential for allowing users to access websites, services, and other resources using easily memorable domain names rather than numerical IP addresses.

DNS Resolution Process:
When a user enters a domain name (e.g., www.example.com) into a web browser, the DNS resolution process begins. The DNS server is responsible for translating the domain name into the corresponding IP address.
The resolution process involves querying multiple DNS servers in a hierarchical manner until the IP address for the domain is obtained.

Types of DNS Servers:

Root DNS Servers: These are the starting point for DNS resolution. They provide information about the authoritative DNS servers for top-level domains.

Top-Level Domain (TLD) DNS Servers: Responsible for specific top-level domains (e.g., .com, .org).

Authoritative DNS Servers: Store and provide information about a specific domain. Each domain has its authoritative DNS servers.

Recursive DNS Servers: Perform the actual work of resolving DNS queries. They may query other DNS servers to obtain the necessary information.

DNS Records:

DNS servers store various types of records associated with domain names. Common DNS record types include:
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AAAA Record: Maps a domain name to an IPv6 address.
CNAME (Canonical Name) Record: Alias of one domain to another.
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NS (Name Server) Record: Specifies authoritative DNS servers for the domain.

DNS Caching:
To improve performance and reduce latency, DNS servers often cache the results of previous DNS queries. Cached information can be reused for subsequent requests, reducing the need to query authoritative DNS servers repeatedly.

Forwarders:
DNS servers can be configured to use forwarders, which are other DNS servers that they forward queries to instead of performing the full resolution process. This can be useful for improving efficiency and reducing the load on root and TLD servers.

DNS Security:
DNS is vulnerable to various attacks, such as DNS spoofing and cache poisoning. DNS Security Extensions (DNSSEC) is a set of extensions designed to add an extra layer of security by signing DNS data cryptographically.

Primary and Secondary DNS Servers:
Organizations often have both primary and secondary DNS servers for redundancy. The primary server is the authoritative server for a domain, and the secondary server replicates the DNS information from the primary server to provide backup in case of failures.

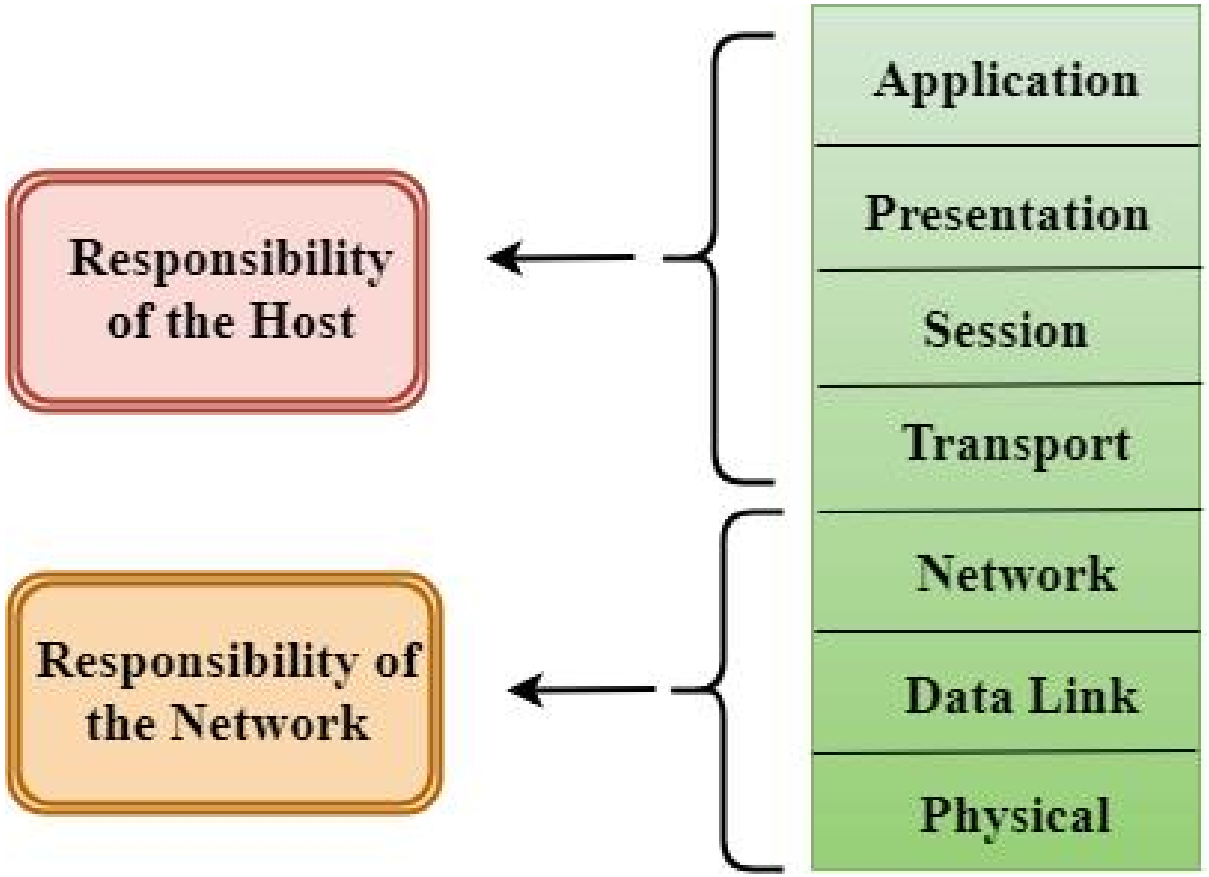
Recursion:

DNS servers can operate in recursive or non-recursive modes. Recursive DNS servers perform the entire DNS resolution process on behalf of the client, while non-recursive servers provide only the information they have cached.

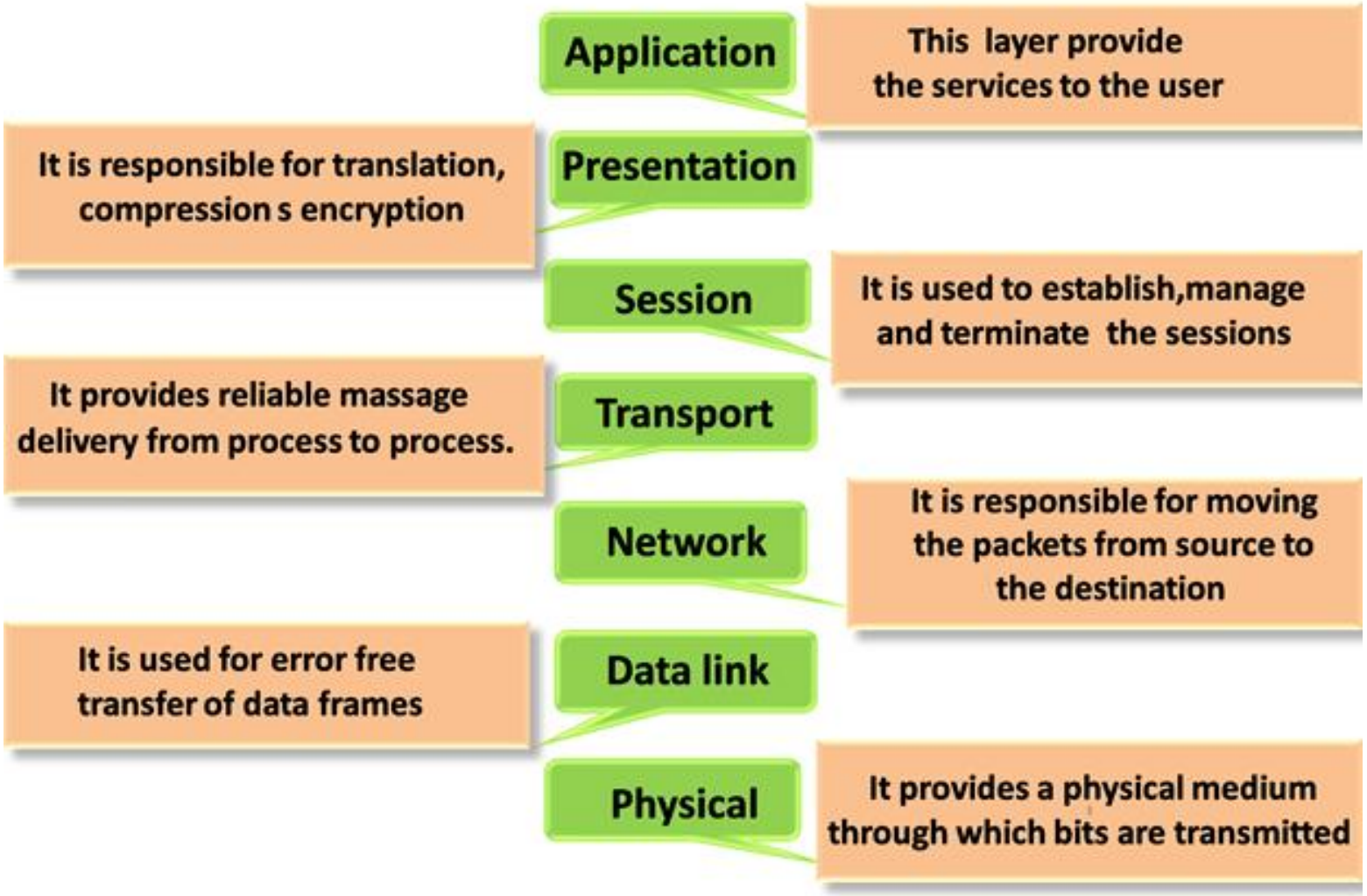
OSI model and TCP/IP

The OSI (Open Systems Interconnection) model and the TCP/IP (Transmission Control Protocol/Internet Protocol) model are conceptual frameworks that standardize the functions of a telecommunication or computing system into layers. While the OSI model has seven layers, the TCP/IP model has four layers. Each layer in both models has a specific set of functions, and they work together to enable communication over a network.

Attribute of OSI Model



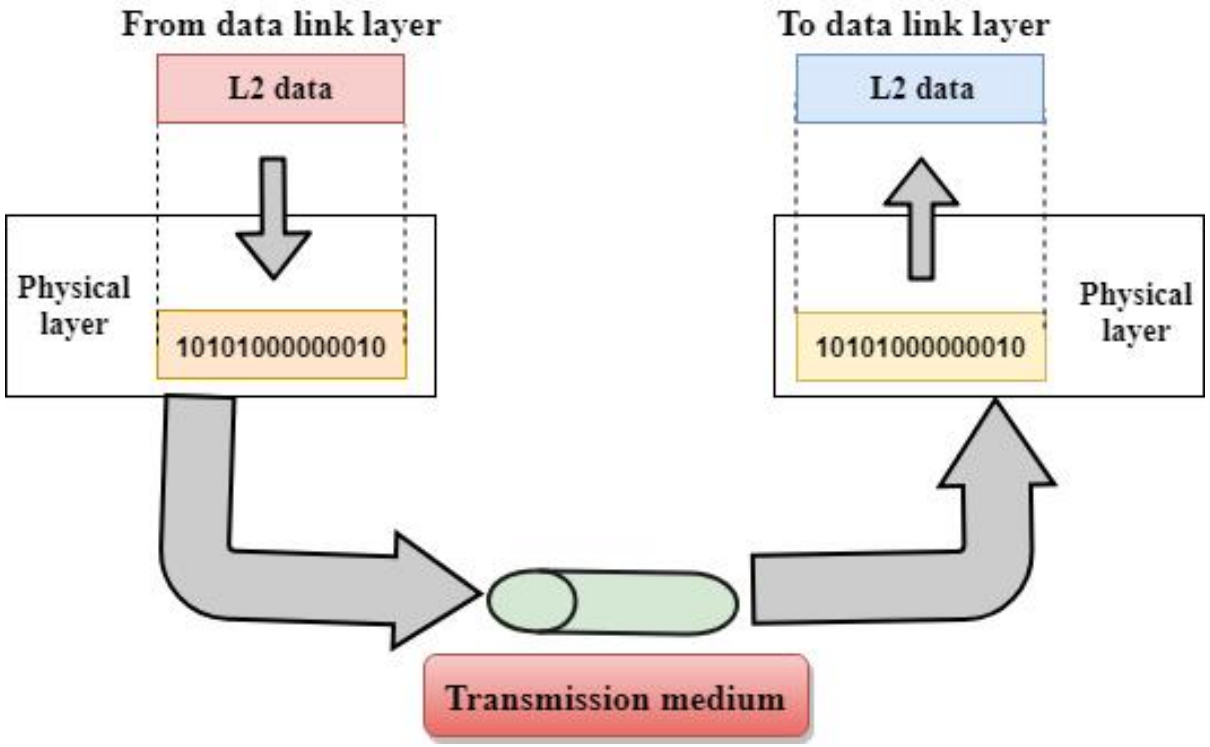
Functions of the OSI Layers



Physical Layer

The main functionality of the physical layer is to transmit the individual bits from one node to another node.
It is the lowest layer of the OSI model.

It establishes, maintains and deactivates the physical connection.
It specifies the mechanical, electrical and procedural network interface specifications.


From data link layer
L2 data
Physical layer
10101000000010
To data link layer
L2 data
Physical layer
10101000000010
Transmission medium

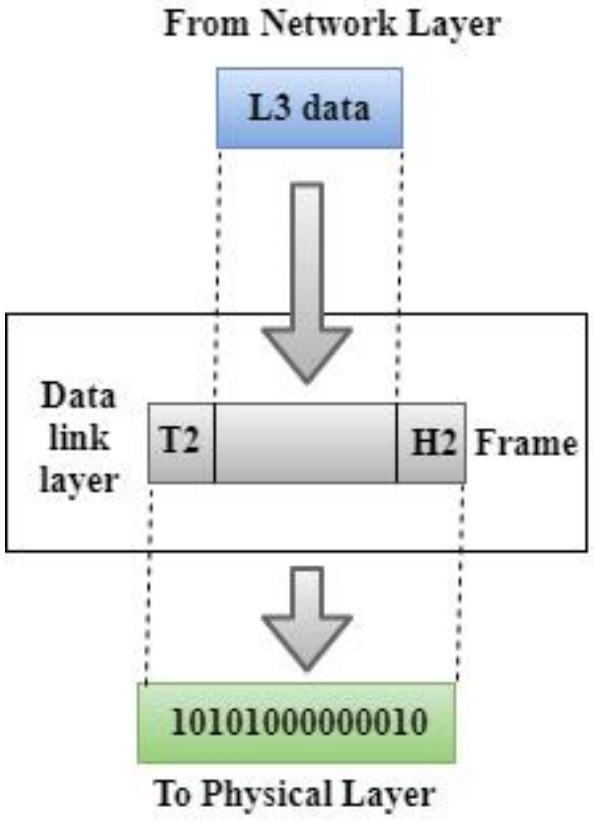
Data-Link Layer

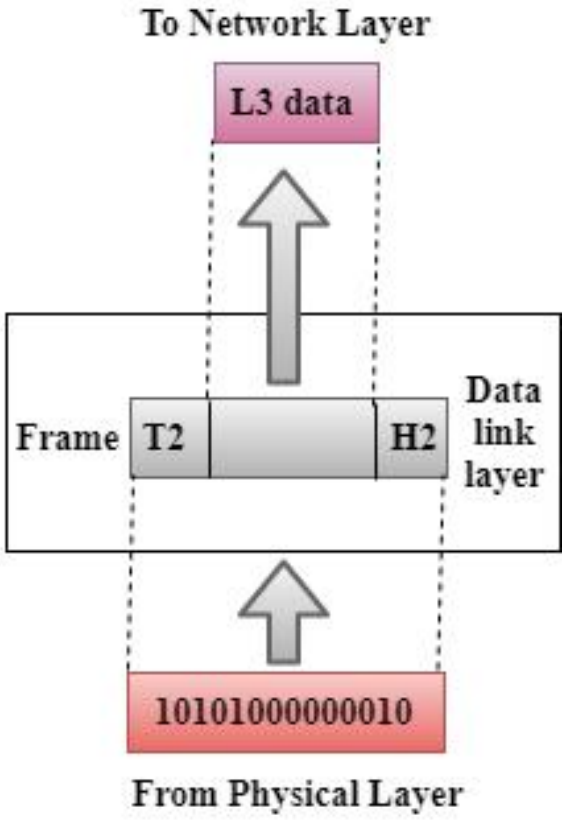
This layer is responsible for the error-free transfer of data frames.

It defines the format of the data on the network.

It provides a reliable and efficient communication between two or more devices.

It is mainly responsible for the unique identification of each device that resides on a local network.

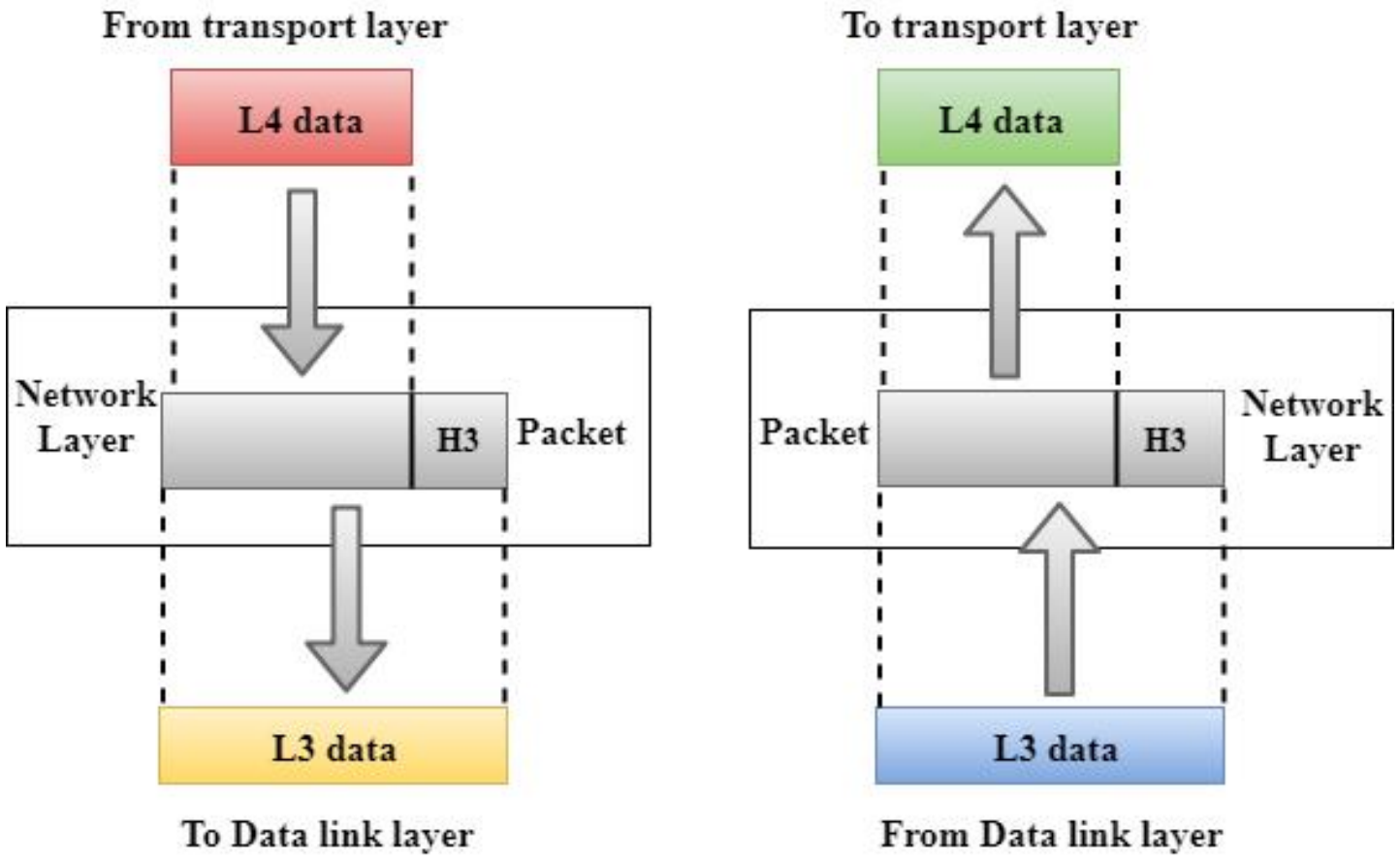

From Network Layer
L3 data
Data link layer
T2
H2
Frame
10101000000010
To Physical Layer


To Network Layer
L3 data
Frame
T2
H2
Data link layer
10101000000010
From Physical Layer

Network layer

Routers are the layer 3 devices, they are specified in this layer and used to provide the routing services within an internetwork(LAN).

The protocols used to route the network traffic are known as Network layer protocols. Examples of protocols are IP and Ipv6.


From transport layer
L4 data
Network Layer
H3
Packet
L3 data
To Data link layer
To transport layer
L4 data
Packet
H3
Network Layer
L3 data
From Data link layer

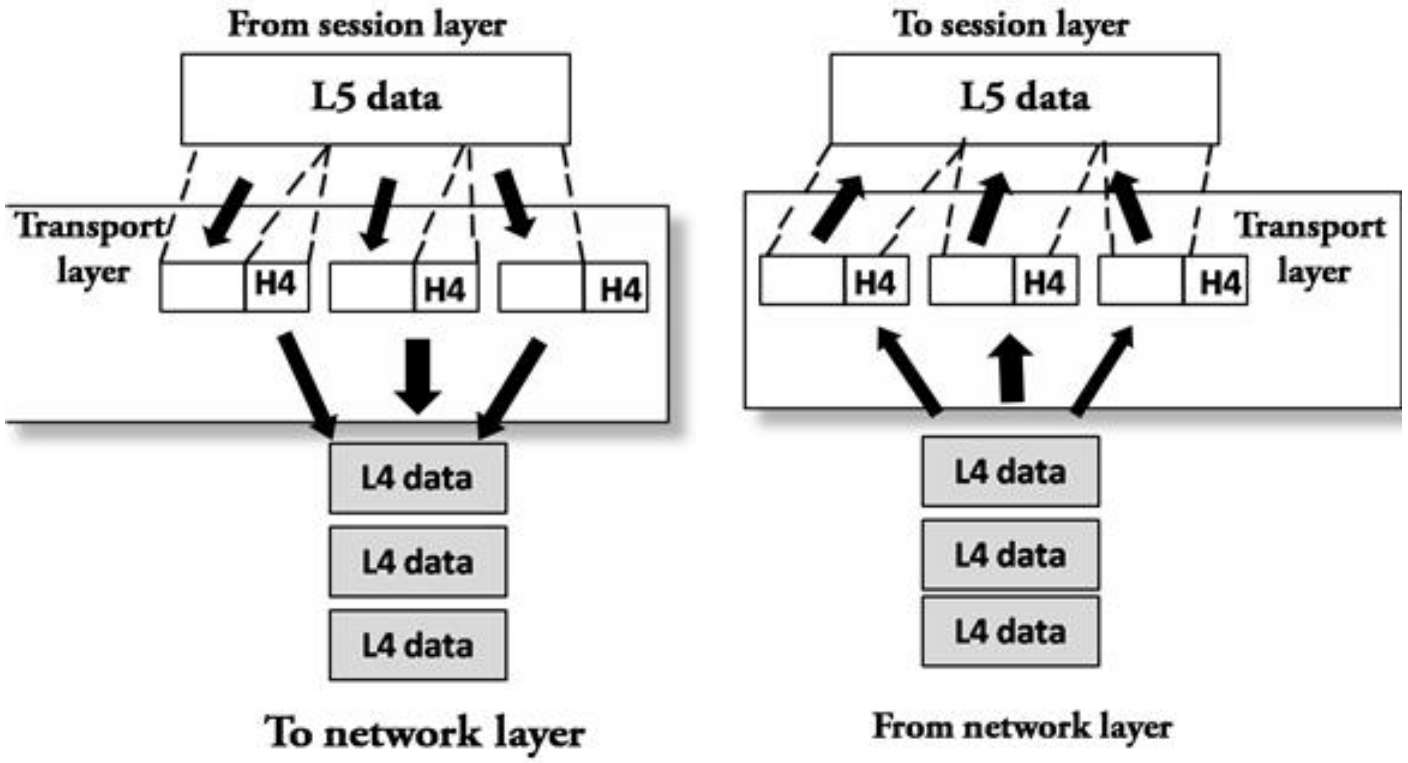
Transport layer

The Transport layer is a Layer 4 ensures that messages are transmitted in the order in which they are sent and there is no duplication of data.

The main responsibility of the transport layer is to transfer the data completely.

It receives the data from the upper layer and converts them into smaller units known as segments.

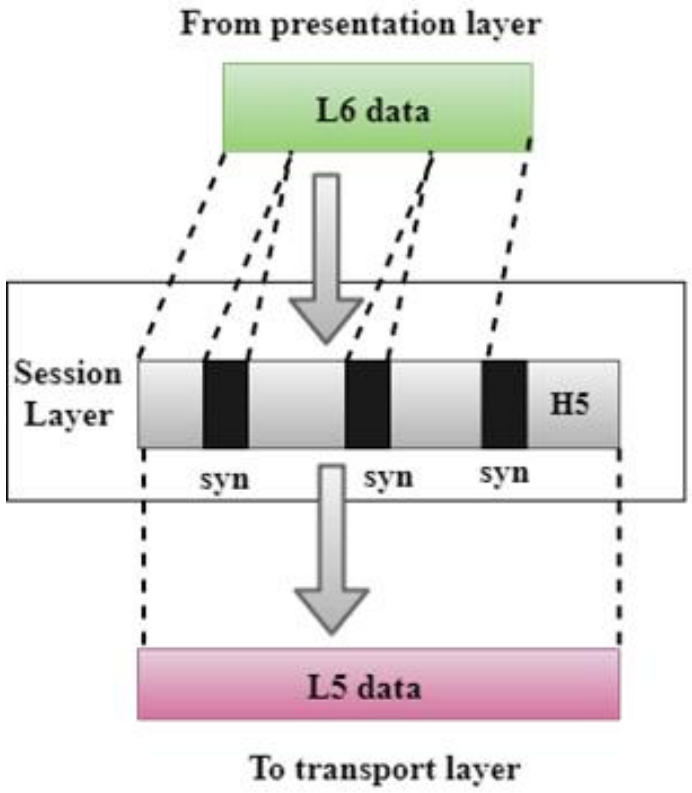
This layer can be termed as an end-to-end layer as it provides a point-to-point connection between source and destination to deliver the data reliably.

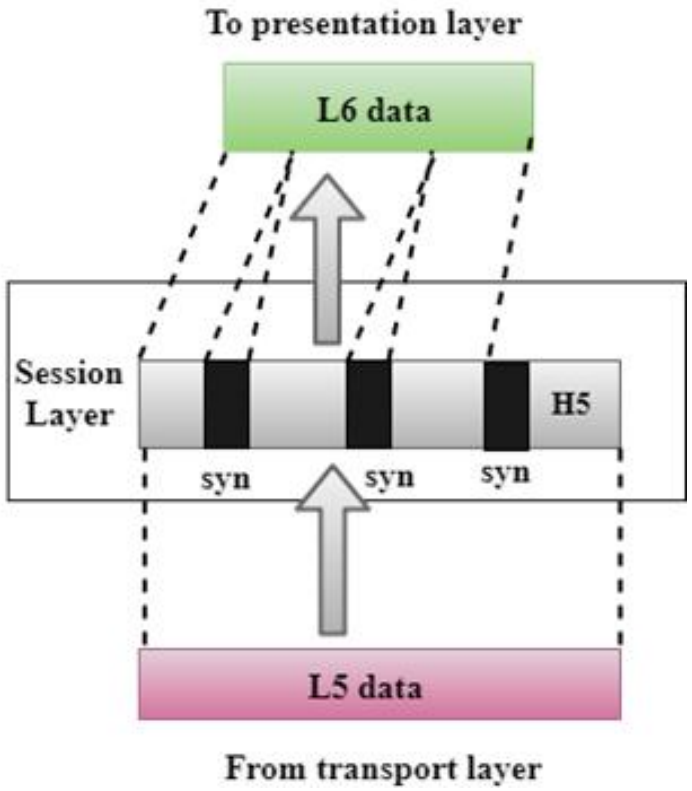

From session layer
L5 data
Transport layer
H4
H4
H4
L4 data
L4 data
L4 data
To network layer
To session layer
L5 data
Transport layer
H4
H4
H4
L4 data
L4 data
L4 data
From network layer

Session layer

It is a layer 5 in the OSI model.

The Session layer is used to establish, maintain and synchronizes the interaction between communicating devices.


From presentation layer
L6 data
Session Layer
H5
syn
syn
syn
L5 data
To transport layer


To presentation layer
L6 data
Session Layer
H5
syn
syn
syn
L5 data
From transport layer

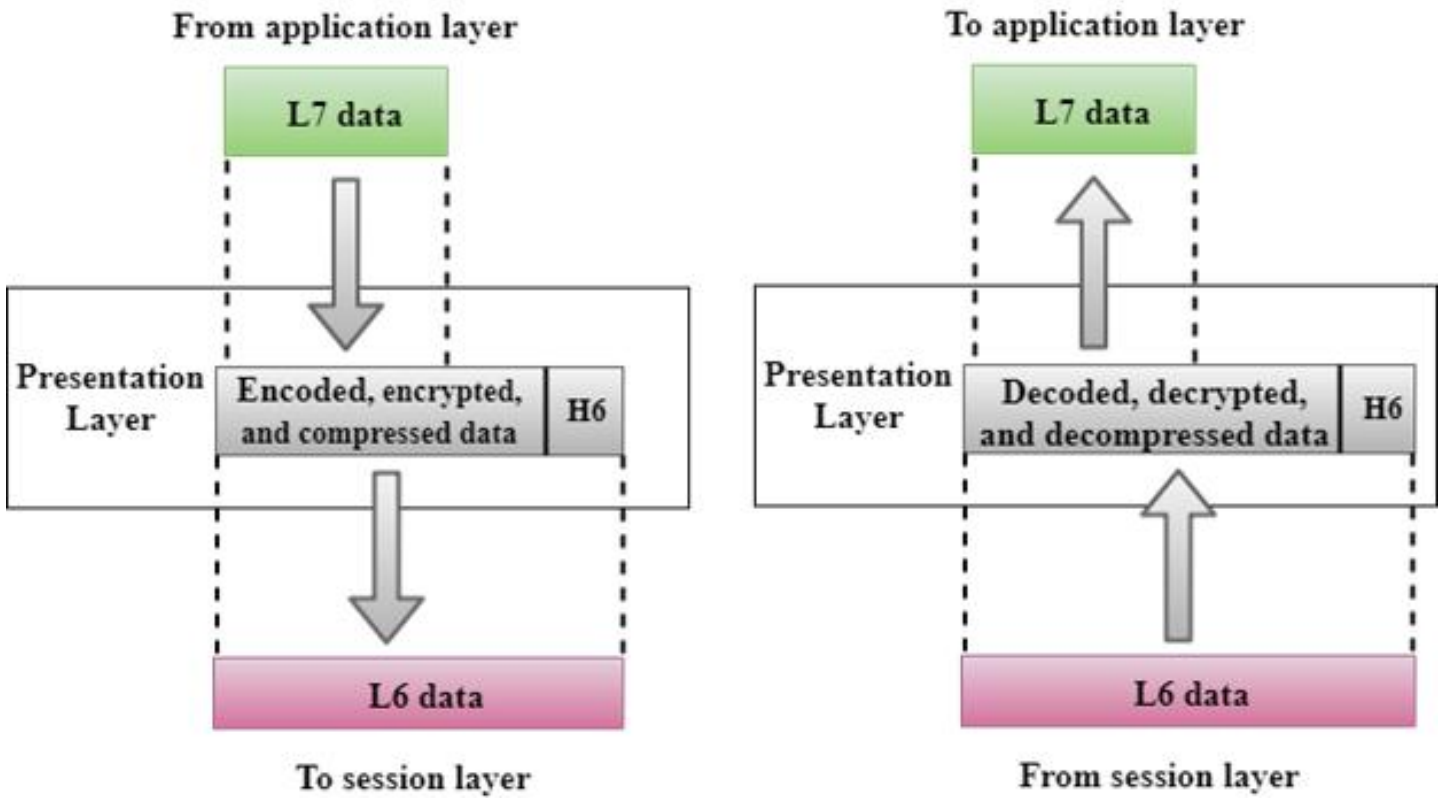
Presentation layer

A Presentation layer is mainly concerned with the syntax and semantics of the information exchanged between the two systems.

It acts as a data translator for a network.

This layer is a part of the operating system that converts the data from one presentation format to another format.

The Presentation layer is also known as the syntax layer.


From application layer
L7 data
Presentation Layer
Encoded, encrypted, and compressed data
H6
L6 data
To session layer
To application layer
L7 data
Presentation Layer
Decoded, decrypted, and decompressed data
H6
L6 data
From session layer

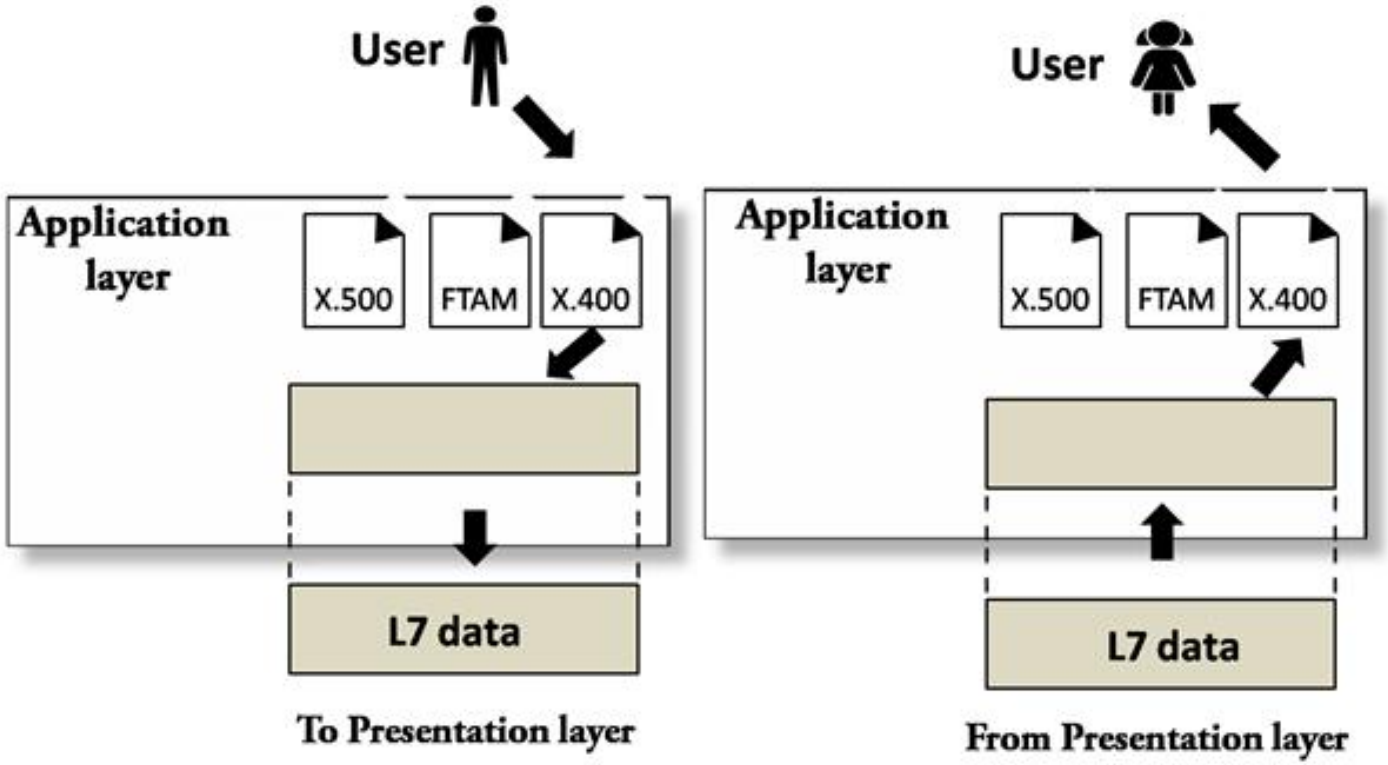
Application layer

An application layer serves as a window for users and application processes to access network service.

It handles issues such as network transparency, resource allocation, etc.

An application layer is not an application, but it performs the application layer functions.

This layer provides the network services to the end-users.


User
Application layer
X.500
FTAM
X.400
L7 data
To Presentation layer
User
Application layer
X.500
FTAM
X.400
L7 data
From Presentation layer

TCP/IP Model

The Internet protocol suite, commonly known as TCP/IP, is a framework for organizing the set of communication protocols used in the Internet and similar computer networks according to functional criteria.

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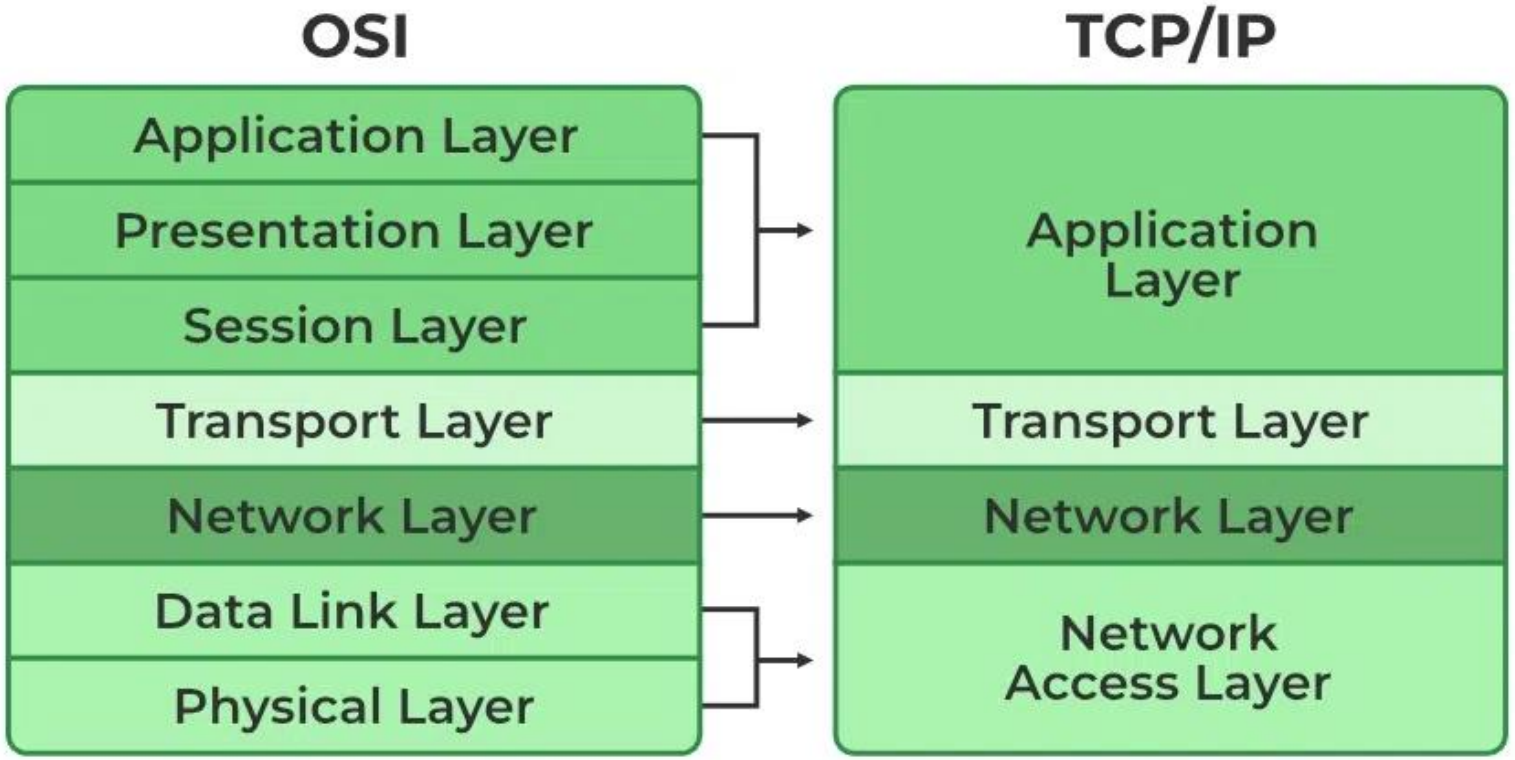
OSI
TCP/IP
Application Layer
Presentation Layer
Session Layer
Transport Layer
Network Layer
Data Link Layer
Physical Layer
Application Layer
Transport Layer
Network Layer
Network Access Layer

Link Layer (or Network Interface Layer):

Combines functionalities of the OSI Physical and Data Link layers.

Concerned with the physical and data link aspects of network communication.

Internet Layer

Corresponds to the OSI Network Layer.
Responsible for logical addressing (IP addresses) and routing.
Includes the IP (Internet Protocol) and ICMP (Internet Control Message Protocol).

Transport Layer

Similar to the OSI Transport Layer.
Manages end-to-end communication, providing reliable or unreliable delivery.
Includes TCP (Transmission Control Protocol) and UDP (User Datagram Protocol).

Application Layer

Similar to the OSI Session, Presentation, and Application layers.

Provides network services directly to end-users or applications.

Includes a variety of application protocols such as HTTP, FTP, and SMTP.